

Quick Start Guide

Sedimentology-GrainSize toolkit

1. Install

Offline (Anaconda or pip):

```
pip install -r requirements.txt
```

Google Colab: open **notebooks/Master_Workflow_Colab.ipynb** and run the first cell - it installs everything automatically.

2. Run the master workflow

The fastest path is three lines of Python:

```
import grainsize as gs
out = gs.run_workflow('sample_data/river.csv', outdir='output')
out['summary_df']
```

This produces, for every sample: histogram, frequency curve, cumulative curve, Passega C-M, Folk & Shepard ternary classification, and a Visher log-probability analysis - plus figures (PNG/TIFF/SVG/PDF), summary CSVs, and PDF + HTML reports.

3. Use your own data

Prepare a CSV with a **sample** column and one column per grain-size class (headers in phi, mm, or um; cells = weight %). Then:

```
out = gs.run_workflow('mydata.csv', size_unit='phi',
    value_type='frequency', outdir='output')
```

4. Where to find outputs

```
output/figures/ - all figures (per sample + dataset diagrams)
output/csv/ - summary and segment tables
output/reports/ - PDF report
output/html/ - HTML report (open in any browser)
```

Example output - histogram:

